

Grade 7 Accelerated
Unit 1
Lesson 8 Practice

Name _____
Date _____
Period _____

Use the table of possible solutions for questions 1–2.

Possible Solutions				
-11.5	36	$6\frac{3}{8}$	-8	-0.1212
-25	-9.5	-7.75	$6\frac{2}{7}$	-17.3

Determine the values from the table that are solutions to the inequalities.

1. $-3 + \frac{x}{2} < -7$

$+3 \quad +3$

$x < -8; -11.5, -25, -9.5, -17.3$

$2 \cdot \frac{x}{2} < -4 \cdot 2$

2. $-3 + \frac{x}{2} > -7$

$+3 \quad +3$

$x > -8; 36, 6\frac{3}{8}, -7.75, 6\frac{2}{7}, -0.1212$

$2 \cdot \frac{x}{2} > -4 \cdot 2$

3. Rewrite $-3 + \frac{x}{2} < -7$ and $-3 + \frac{x}{2} > -7$ so that -8 is a solution.

Explain your reasoning.

Change $<$ and $>$ to $\leq$ and $\geq$ because greater than/less than or equal to means -8 will be included.

4. Inequalities have infinite solutions that are represented on the number line.

5. Tanner and and Jonah were solving the inequality $-3 - \frac{y}{7} < -5$.

Jonah said the solution is $y < -14$, while Tanner said the solution is $y > -14$. Who do you agree with or do you agree with neither? Justify your reasoning mathematically.

$-3 - \frac{y}{7} < -5$

$-7 \cdot \frac{-y}{7} < -2 \cdot -7$

$y > 14$

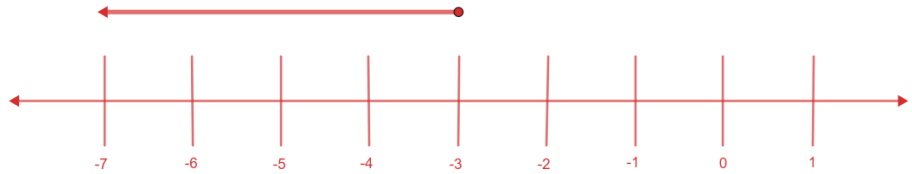
$+3 \quad +3$

Neither because $-2 \cdot -7 = +14$

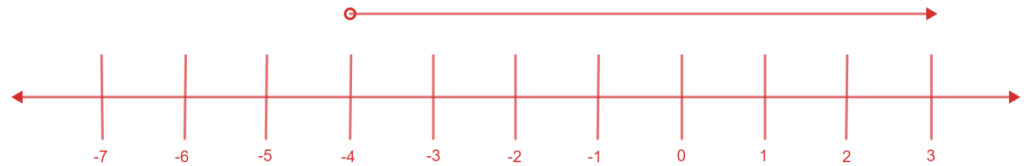
6. When multiplying or dividing by negative coefficient in an inequality, the resulting inequality symbol will change to its opposite.

For questions 7–10, solve each inequality and graph the solution set.

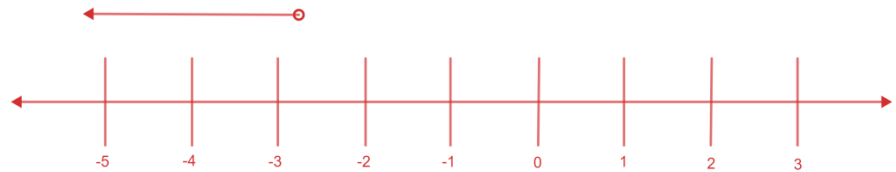
7. $-8 \leq \frac{a}{-3} - 9$
 $+9 \quad +9$
 $-3 \cdot 1 \leq \frac{a}{-3} \cdot -3$
 $-3 \geq a \text{ or } a \leq -3$



8. $9m + 16 > -20$
 $-16 \quad -16$
 $\frac{9m}{9} > \frac{-36}{9}$
 $m > -4$



9. $2.8 < -1.3p - 0.5$
 $+0.5 \quad +0.5$
 $\frac{3.3}{-1.3} < \frac{-1.3p}{-1.3}$
 $-2\frac{7}{13} > p \text{ or } p < -2\frac{7}{13}$



10. $-2 + \frac{5}{4}x \geq -12$
 $+2 \quad +2$
 $\frac{4}{5} \cdot \frac{5}{4}x \geq -10 \cdot \frac{4}{5}$
 $x \geq -8$

